《计算机图形学》课程简介

夏时洪

中国科学院计算技术研究所

中国科学院大学计算机学院

2026-03-05

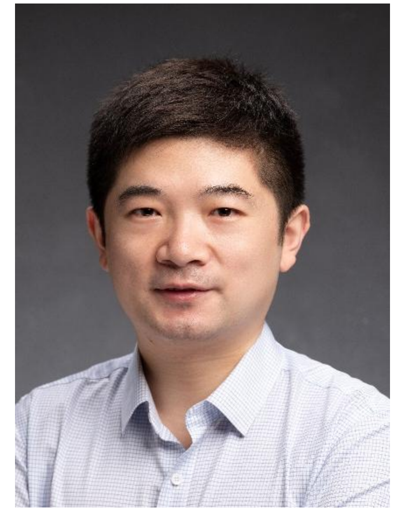
提纲

1. 关于本课
2. 计算机图形学早期发展回顾
3. 对计算机图形学的再认识
4. AI4CG？ CG4AI？

Part 1: 关于本课

教学团队

夏时洪 博士



高林 博士

教师助教：周洁

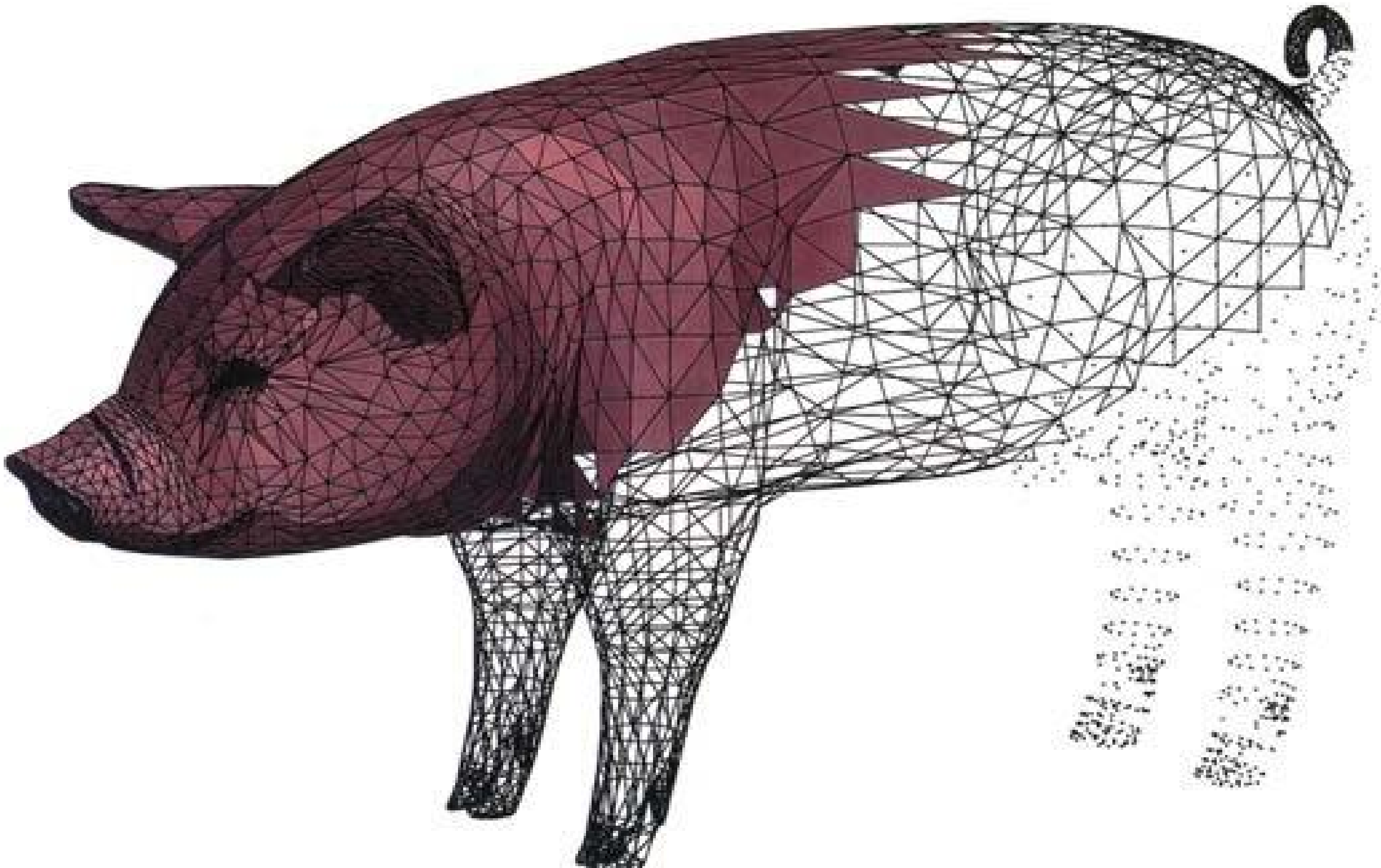
教学目的

本课程为计算机、电子、自动化等学科领域研究生的专业普及课，主要讲授计算机图形学领域的主要概念、算法思想和基本技术。课程内容涉及几何计算、渲染、动画模拟、三维重建与运动捕捉等。通过本课程的学习，希望学生能掌握计算机图形学的基本概念和经典算法，了解计算机图形学的前沿研究领域和最新研究成果，通过课程实验设计对所学习的内容进行实践验证。

教学内容

第一章 课程简介
第二章 基本知识
第三章 三角网格
第四章 网格变形
第五章 光线跟踪
第六章 纹理
第七章 阴影
第八章 材质反射属性模型(BRDF)
第九章 非真实感绘制
第十章 计算机动画
第十一章 人脸动画
第十二章 三维重建
第十三章 数字几何处理
第十四章 运动合成基础
第十五章 运动捕捉基础

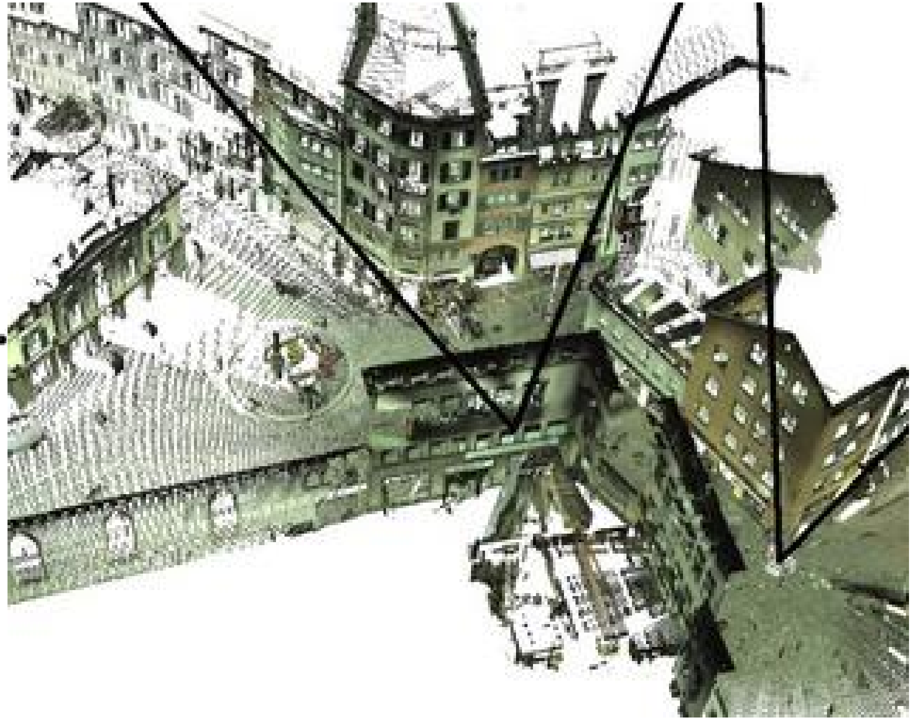
几何计算



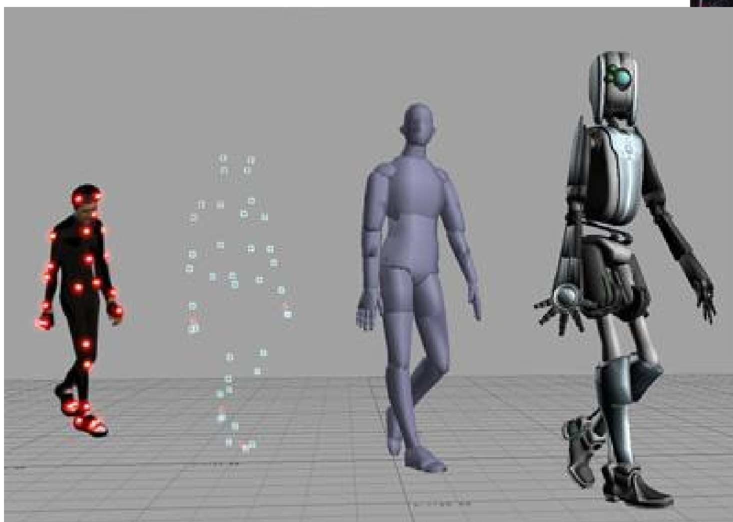
渲染

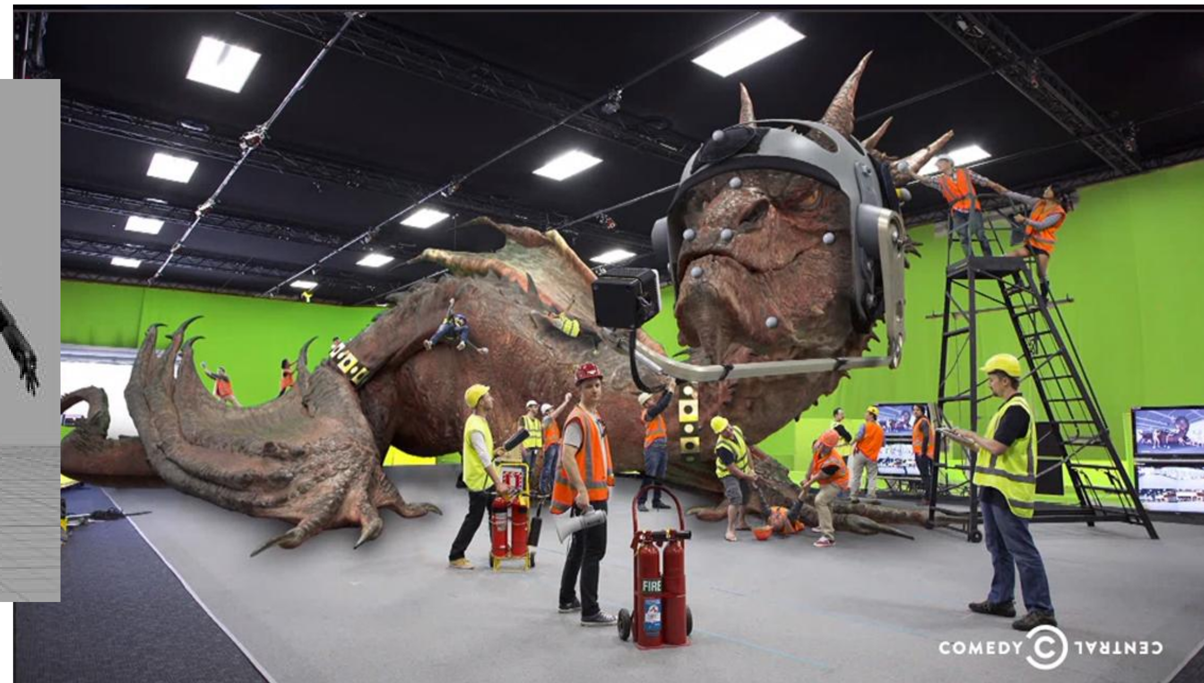
动画模拟

三维重建



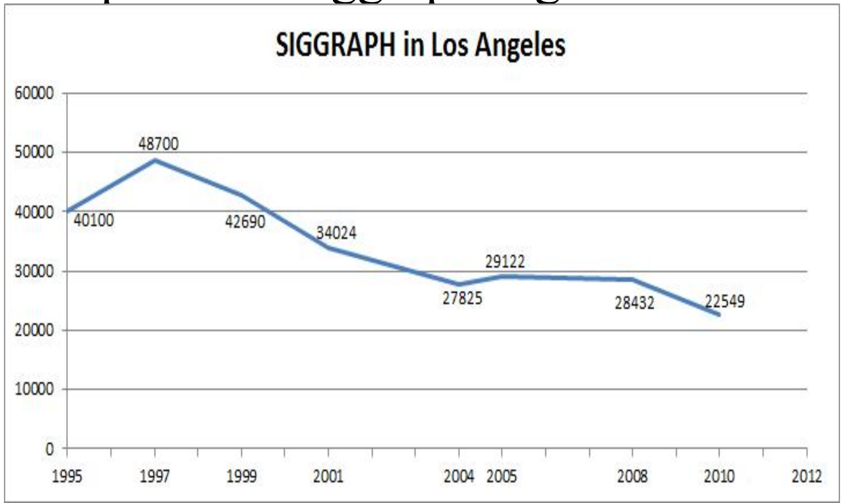
运动捕捉





参考资料（1/4）

- ACM SIGGRAPH全称 “the Special Interest Group on Computer Graphics and Interactive Techniques”
- 60年代中期，由Brown 大学的教授Andries van Dam (Andy) 和IBM公司的Sam Matsa发起
- 1974年，在Colorado大学召开了第一届SIGGRAPH年会
- 2012， Los Angeles http://www.siggraph.org 21000+ attendee



参考资料（2/4）

Top Conference/Journal

- ACM SIGGRAPH
- ACM SIGGRAPH Asia
- ACM TOG

Graphics Lab

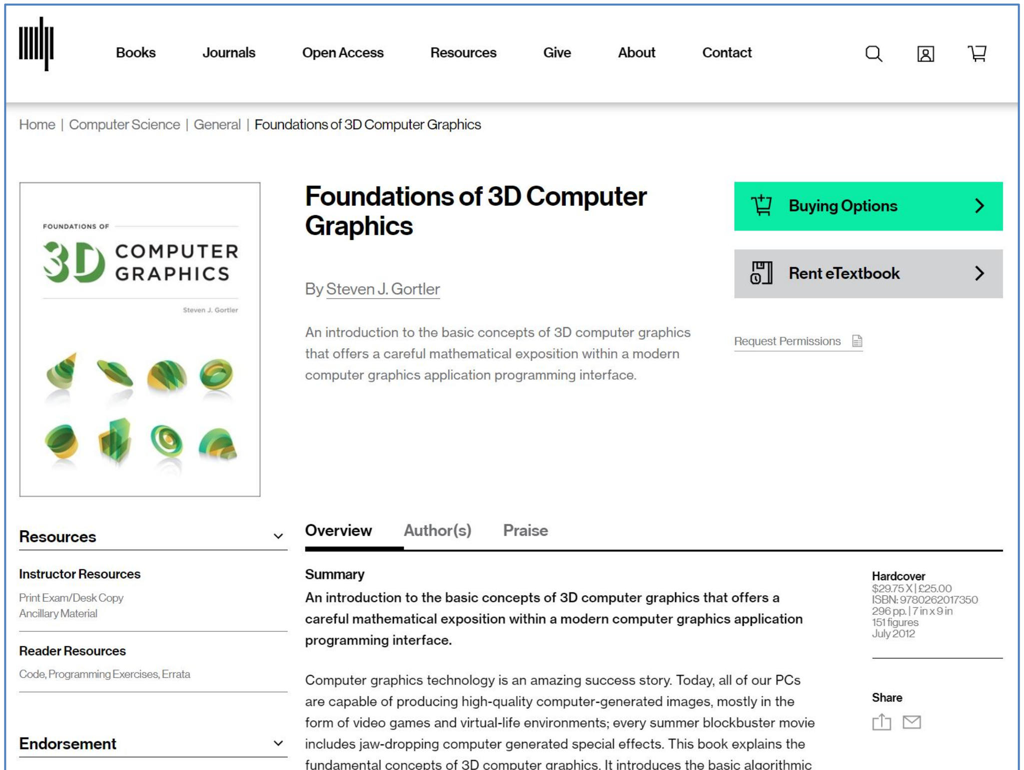
- Graphics Group at Stanford University
- Graphics Lab at MIT
- Graphics Research at Caltech
- Graphics Group at UC Berkeley
- Graphics Group at Cornell University
- Graphics Lab at University of Washington
- Graphics Group at Brown University
- Graphics Lab at Carnegie Mellon University

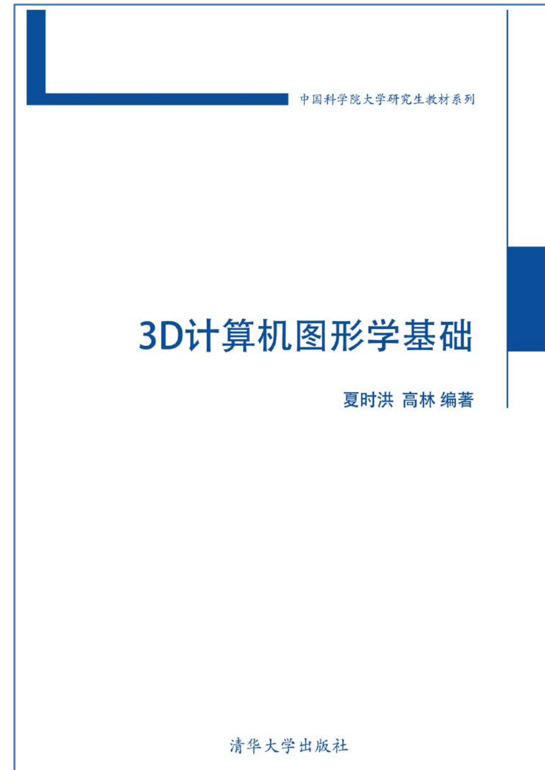
参考资料（3/4）

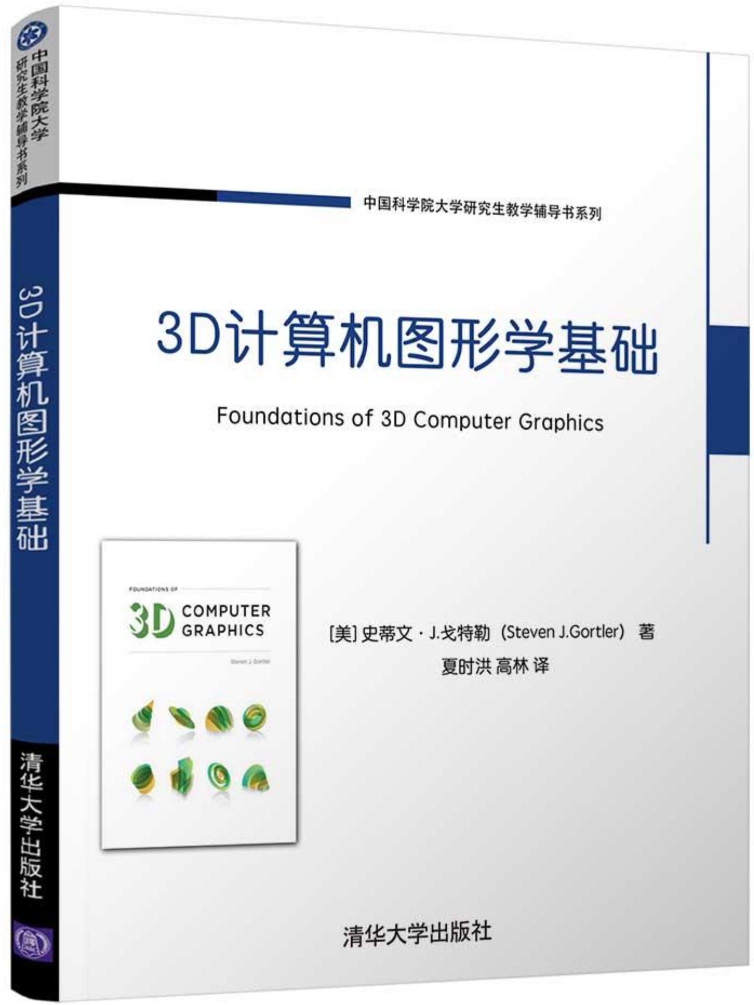
- http://kesen.realtimerendering.com/
- https://dblp.uni-trier.de/db/conf/siggraph/index.html
- https://dl.acm.org/loi/tog
- https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=2945

参考资料（4/4）

- https://mitpress.mit.edu/books/foundations-3d-computer-graphics







[美] 史蒂夫·J.戈特勒(Steven J.Gortler)著; 夏时洪, 高林译. 3D计算机图形学基础. 清华大学出版社. ISBN: 978-7-302-56105-7. 2020.

小结

计算机图形学研究领域顶级会议和顶级刊物是什么?

如何找到感兴趣的计算机图形学学术论文?

计算机图形学领域有哪些研究内容?

中科院计算所 INSTITUTE OF COMPUTING TECHNOLOGY

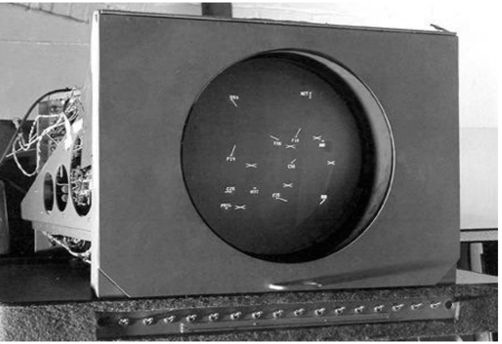
Part 2: 计算机图形学早期发展回顾

50年代

– 1950年，第一台图形显示器作为美国麻省理工学院（MIT）旋风I号（Whirlwind I） 计算机的附件诞生。

– 1958年，美国Calcomp公司由联机的数字记 录仪发展成滚筒式绘图仪，GerBer公司把 数控机床发展成为平板式绘图仪。

– 50年代末期，MIT的林肯实验室在“旋风”计算机上开发 SAGE 空中防御体系。

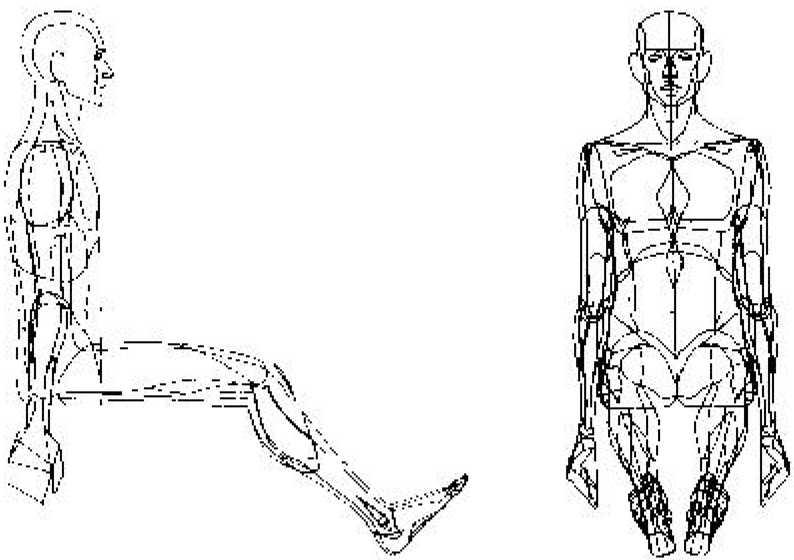


左：Whirlwind I computer, 1951

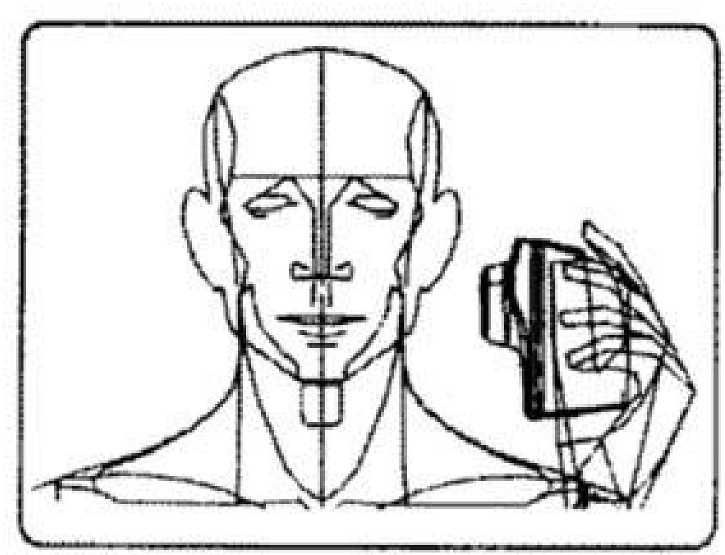
右：User interacting with a SAGE monitor with a light pen, 1958

60年代

– 1960年: William Fetter (-2002)首次使用图形来描述其在波音公司所从事的工作。



Fetter's Boeing Man (1964)



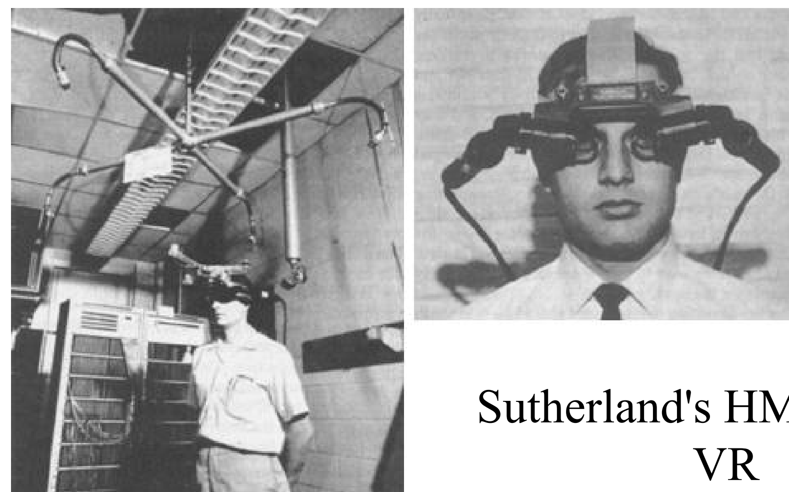
Ad for Norelco by Fetter (1970)

60年代

– 1963年，MIT林肯实验室的 I. E. Sutherland 发表了一篇题为“Sketchpad：一个人机交互通信的图形系统”的博士论文。

– 提出图形学的概念，成就“图形学之父“的英名

– 获图灵奖（1988），IEEE计算机杰出成就奖, Coons奖



Sutherland's HMD(1968)
VR

60年代

– I. E. Sutherland，1938年生，美国人，CMU学士、 Caltech硕士、MIT博士，1963年博士毕业。

– 1964年，26岁时担任国防部高级研究计划署(DARPA)信息处理技术局局长。

– 1965年起，先后在Harvard、Utah、Caltech工作。

– 培养的学生：Bob Sproull，虚拟现实的发明人，第一本图形学教科书，美国工程院院士、美国艺术与科学院院士。(Harvard)

60年代

– Danny Cohen，发明Cohen-Sutherland算法，美国工程院院士，在ARPANet上开发飞行模拟器。(Harvard)

– Alan Kay，发明面向对象编程、窗口图形用户界面； 美国工程院院士、美国艺术与科学院院士，获图灵奖（2003）。 (Utah)

– Edwin Catmull，细分算法发明人，Z-buffer算法发明人， Coons奖，Oscar 奖，曾任Pixar 动画、Disney 动画总裁，图灵奖（2019）。 (Utah)

– Henri Gouraud，Gouraud shading的发明人。 (Utah)

– Frank Crow，图形反走样算法的发明人。

60年代

– 1964 MIT的教授Steven A. Coons提出了被后人称为超限插值的新思想

– 同在60年代早期，法国雷诺汽车公司的工程师Pierre Bézier发展了一套被后人称为Bézier曲线、曲面的理论，成功地用于几何外形设计，并开发了用于汽车外形设计的UNISURF系统。

– Coons方法和Bézier方法是CAGD最早的开创性工作。值得一提的是，计算机图形学的最高奖是以Coons的名字命名的，而获得第一届（1983）和第二届(1985) Steven A. Coons 奖的，恰好是Ivan E. Sutherland和Pierre Bézier，这也算是计算机图形学的一段佳话。

70年代

1971: Henri Gouraud – Gouraud shading.

1972: Nolan Kay Bushnell – Pong, video arcade game

1973: John Whitney. Jr. and Gary Demos –“Westworld”,

first film with computer graphics

1974: Edwin Catmull – texture mapping and Z-buffer hidden-surface algorithm

James Blinn – curved surfaces, refinement of texture mapping

Phone Bui-Toung – specular highlighting

1975: Martin Newell – famous CG teapot, using Bezier patches

Benoit Mandelbrot – fractal/fractional dimension "A Theory of Fractal Sets."

1976: James Blinn – environment mapping and bump mapping

1979: George Lucas formed the new computer graphics division of Lucas film, another major milestone in the history of computer graphics

Utah Teapot

80年代

1980:Turner Whitted – Ray Tracing

1981:Carpenter wrote the first renderer for Lucas film, called REYES

1982:

Steven Lisberger – “Tron”, first Disney movie which makes extensive use of 3-D computer graphics

Tom Brighman – “Morphing”, first film sequence plays a female character which deforms and transforms herself into the shape of a lynx.

John Walkner and Dan Drake – AutoCAD

1983:Jaron Lanier – “DataGlove”, a virtual reality film features a glove installed with switches and sensors to detect hand motion. “virtual reality”

Morphing

DataGlove

80年代

1984:Cindy Goral, Don Greenberg and others at Cornell University –"Modeling the Interaction of Light Between Diffuse Surfaces." Radiosity

1985:Pixar Animation Studios – “Luxo Jr.”

1986:J. Kajiya. The Rendering Equation. Proc. of SIGGRAPH '86

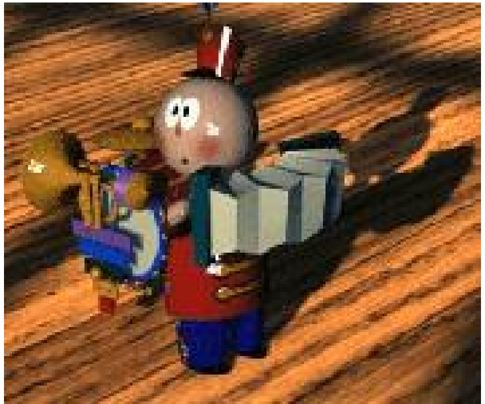
1988:Hanrahan – Renderman， 获图灵奖（2019）

1989:The Pixar Animation Group made history on March 29, 1989 by winning an Oscar at the Academy Awards for their animated short film，，"Tin Toy."

Jaron Lanier coins the phrase Virtual Reality and creates the first commercial business around virtual worlds.

Luxo Jr

Tin Toy



90年代

1990:

Pixar used their new Photorealistic Render software, Renderman, to produce the famous "Shutterbug" image.

Tom Caudell coins the phrase Augmented Reality while at Boeing helping workers assemble cables into aircraft.

1991:

ILM – Terminator 2: Judgment Day (Oscar winner)， the first major digital character to be used in a film

Disney and Pixar – “Beauty and the Beast”, CGI was widely used, Renderman systems provides fast, accurate and high quality digital computer effects.

Shutterbug Rendered at Pixar using Renderman Software

Industrial Light and Magic: Terminator 2: Judgment Day

90年代

1992: Silicon Graphics – OpenGL specification

1993:

University of Illinois -- Mosaic, first graphic Web browser

Steven Spielberg – “Jurassic Park” a successful CG fiction film.

1995:

Buena Vista Pictures – “Toy Story”, first full-length, computer-generated, feature film

NVIDIA Corporation – GeForce 256, GeForce3(2001)

Industrial Light and Magic:

Jurassic Park

Digital Domain:

Titanic

小结

计算机图形学是怎么发展起来的?

计算机图形学有哪些代表性应用?

计算机图形学领域哪些工作获得了图灵奖?

Part 3: 对计算机图形学的再认识

计算机图形学 (1/5)

计算机图形学是利用计算机研究图形的表示、生成、处理、显示的学科，计算机科学中，最为活跃、得到广泛应用的分支之一。

- The representation and manipulation of pictorial data by a computer
- the various technologies used to create and manipulate such pictorial data
- the images so produced, and
- the sub-field of computer science which studies methods for digitally synthesizing and manipulating visual content.

计算机图形学 (2/5)

- Computer generated imagery can be categorized into several different types: two dimensional (2D), three dimensional (3D), and animated graphics.

- 2D computer graphics is the computer-based generation of digital images—mostly from two-dimensional models (such as 2D geometric models, text, and digital images) and by techniques specific to them.

- 2D computer graphics started in the 1950s, based on vector graphics devices. These were largely supplanted by raster-based devices in the following decades.

计算机图形学 (3/5)

•Computer generated imagery can be categorized into several different types: two dimensional (2D), three dimensional (3D), and animated graphics.

•3D computer graphics (in contrast to 2D computer graphics) are graphics that use a three-dimensional representation of geometric data (often Cartesian) that is stored in the computer for the purposes of performing calculations and rendering 2D images. Such images may be stored for viewing later or displayed in real-time.

•3D computer graphics are often referred to as 3D models.

计算机图形学 (4/5)

- Rendering = constructing 2D images from 3D models
- Modeling = representing 3D objects
- Animation = simulating changes over time

计算机图形学 (5/5)

- image/video editing
- 3D printing
- VR
- AR/MR

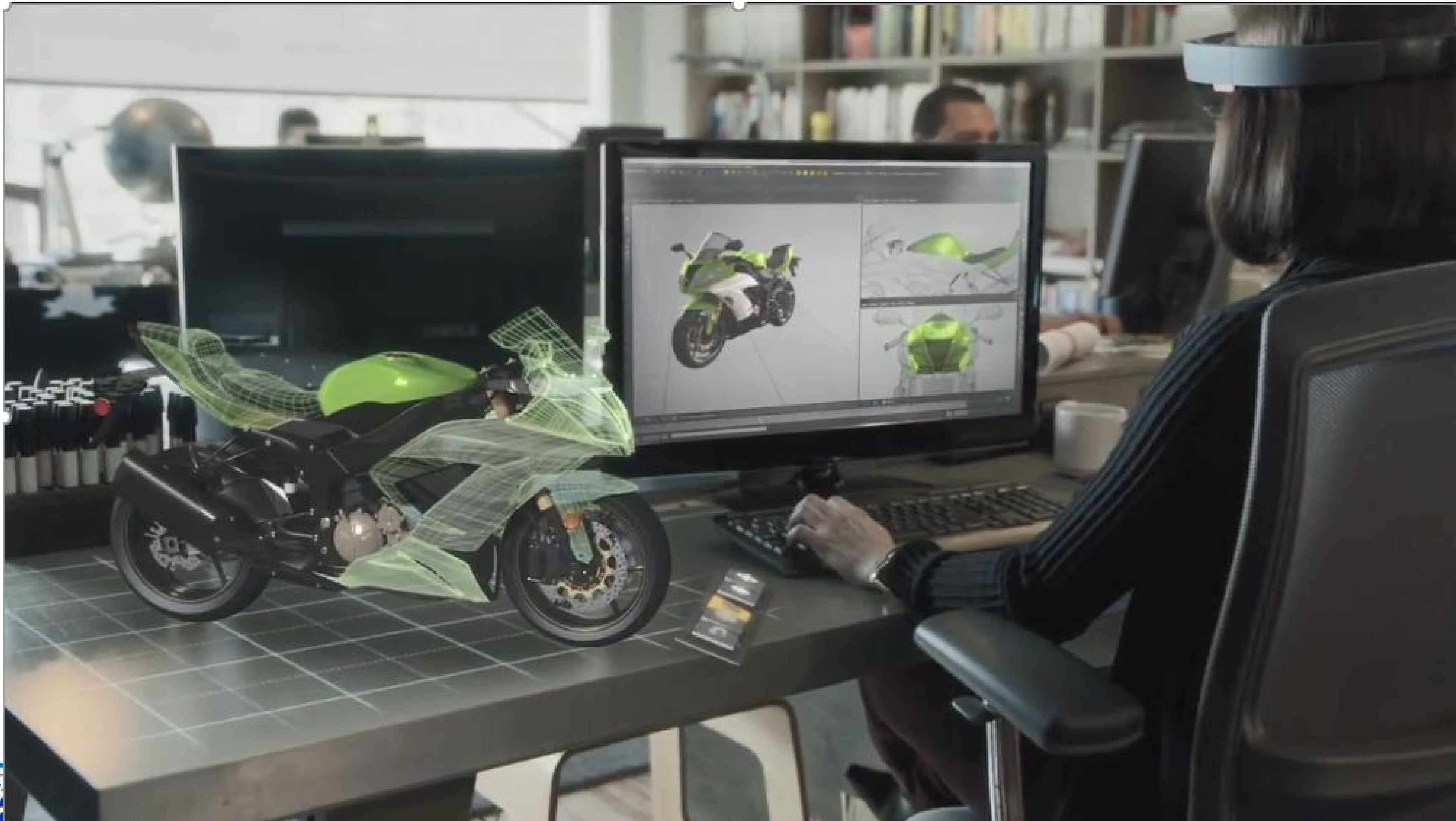
应用领域

- **Entertainment**
- **Computer-aided design**
- **Scientific visualization**
- **Training**
- **Education**
- **E-commerce**
- **Computer art**
- **Medical Imaging**
- **Virtual reality**
- **3D print**
- **……**

电影制作

军事仿真

HoloLens



小结

计算机图形学研究内容不断演变，保持活力；

计算机图形学还会在哪些领域发挥作用?

是否会诞生新的标志性应用?

Part 4: AI4CG? CG4AI?

AI4CG? CG4AI?

- Rendering
- Modeling
- Animation
- Imaging/Video editing

- NERF, 3D Gaussian Splatting
- G-Deeplearning, PointNet
- DeepMotionEditing, text2motion
- DeepFake, Video Generation

NERF

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

Ben Mildenhall[1⋆] Pratul P. Srinivasan[1⋆] Matthew Tancik[1⋆]
Jonathan T. Barron[2] Ravi Ramamoorthi[3] Ren Ng[1]

[1]UC Berkeley [2]Google Research [3]UC San Diego

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis,
Ben Mildenhall*, Pratul Srinivasan*, Matthew Tancik*, Jonathan Barron, Ravi Ramamoorthi, Ren Ng, ECCV 2020.

NERF

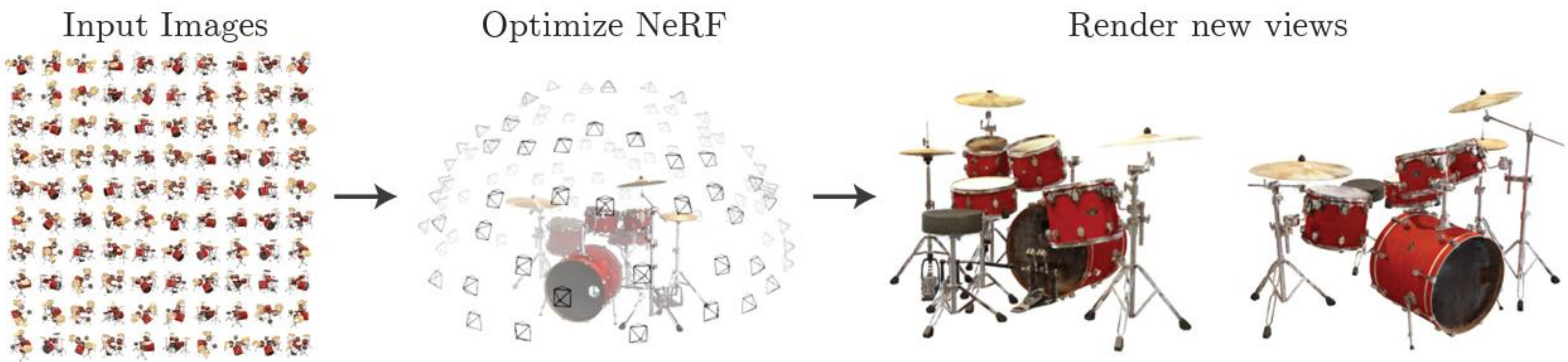


Fig. 1: We present a method that optimizes a continuous 5D neural radiance field representation (volume density and view-dependent color at any continuous location) of a scene from a set of input images. We use techniques from volume rendering to accumulate samples of this scene representation along rays to render the scene from any viewpoint. Here, we visualize the set of 100 input views of the synthetic *Drums* scene randomly captured on a surrounding hemisphere, and we show two novel views rendered from our optimized NeRF representation.

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis,
Ben Mildenhall, *Pratul Srinivasan*, Matthew Tancik*, Jonathan Barron, Ravi Ramamoorthi, Ren Ng, ECCV 2020.

NERF

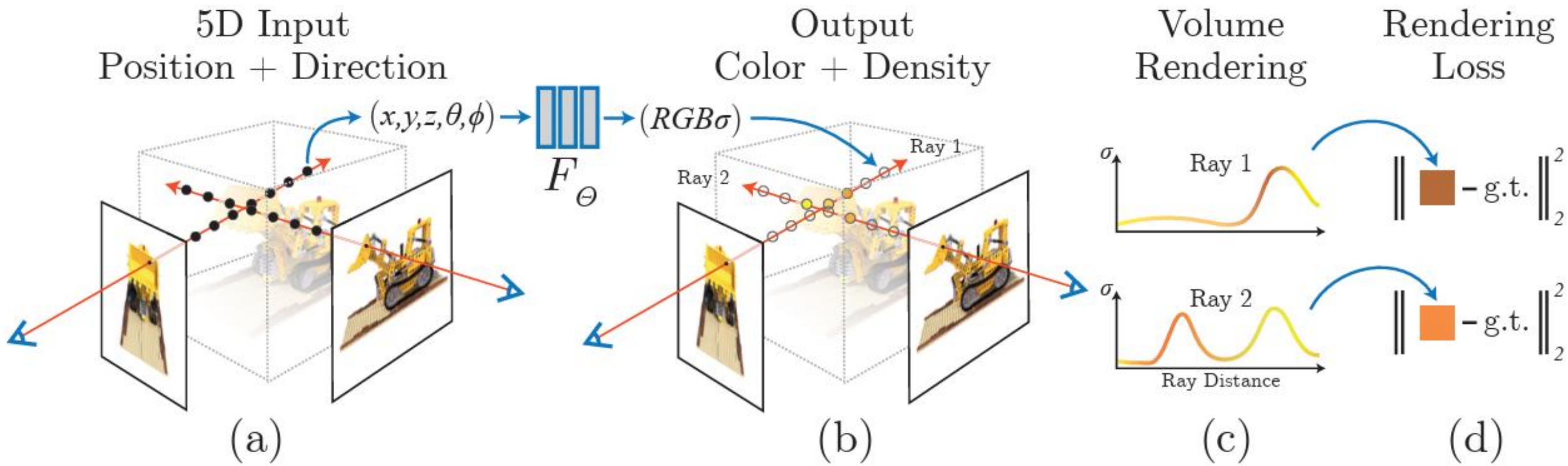
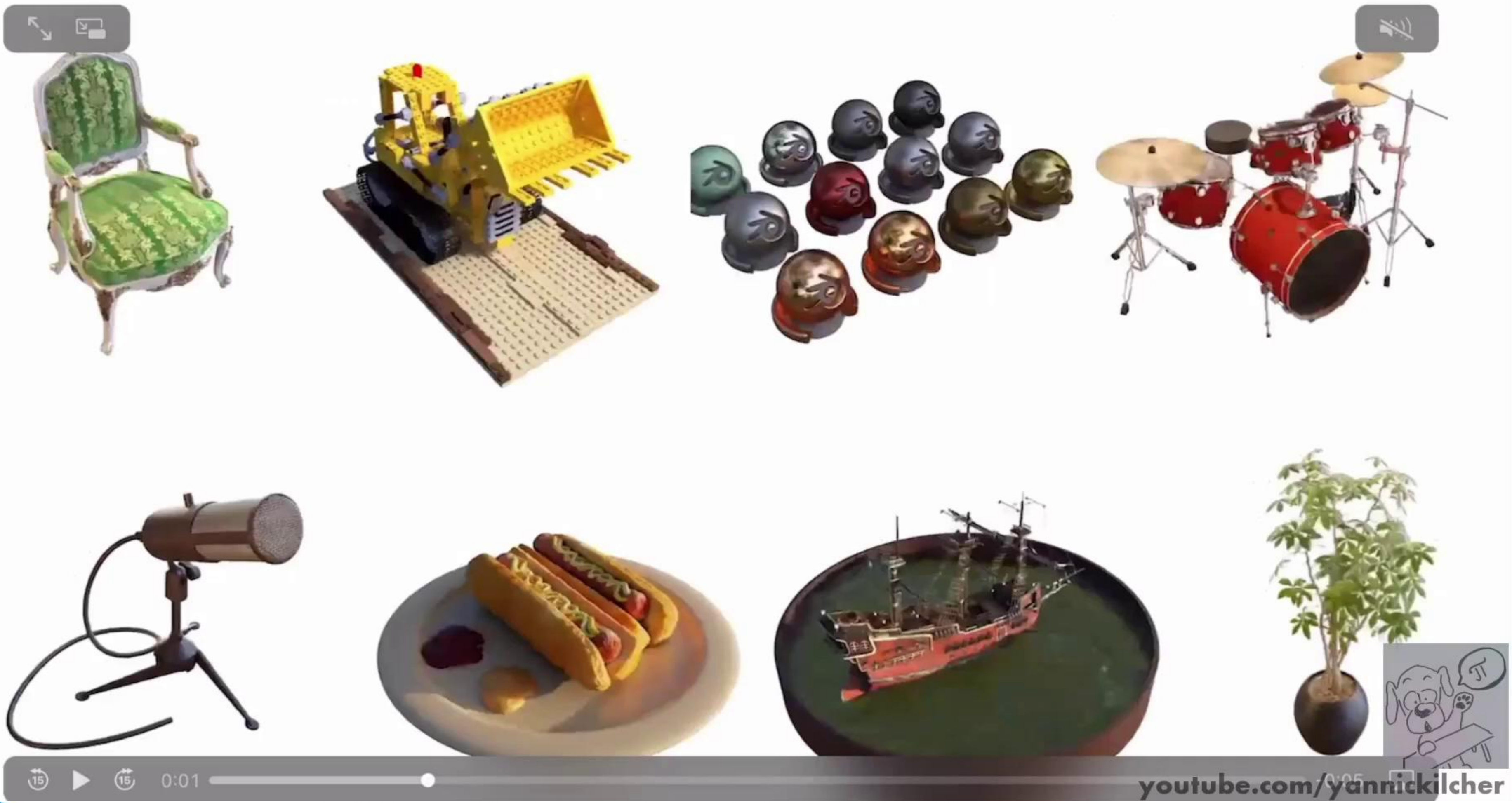


Fig. 2: An overview of our neural radiance field scene representation and differentiable rendering procedure. We synthesize images by sampling 5D coordinates (location and viewing direction) along camera rays (a), feeding those locations into an MLP to produce a color and volume density (b), and using volume rendering techniques to composite these values into an image (c). This rendering function is differentiable, so we can optimize our scene representation by minimizing the residual between synthesized and ground truth observed images (d).

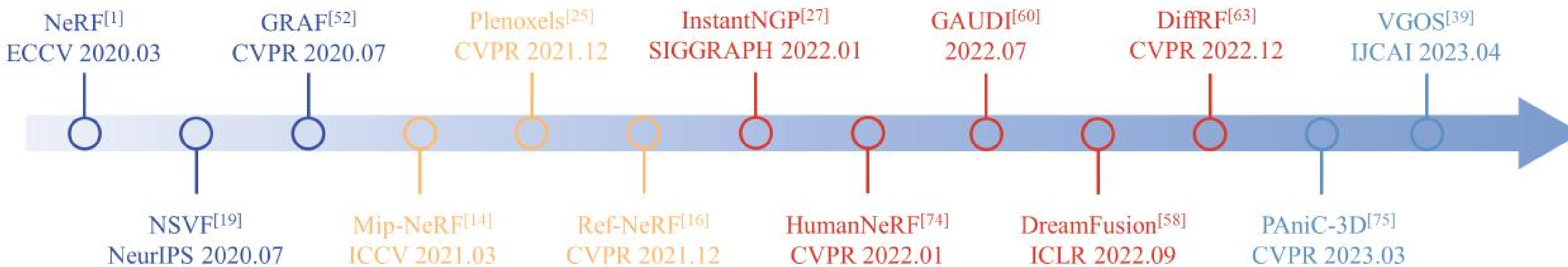
NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis,
Ben Mildenhall, *Pratul Srinivasan*, Matthew Tancik*, Jonathan Barron, Ravi Ramamoorthi, Ren Ng, ECCV 2020.

Here are results on our synthetic dataset of pathtraced objects with realistic non-Lambertian materials. The dataset will be released soon.



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis,
Ben Mildenhall, *Pratul Srinivasan*, Matthew Tancik*, Jonathan Barron, Ravi Ramamoorthi, Ren Ng, ECCV 2020.

NERF演进工作



NeRF[1]
ECCV 2020.03
GRAF[52]
CVPR 2020.07
Plenoxels[25]
CVPR 2021.12
InstantNGP[27]
SIGGRAPH 2022.01
GAUDI[60]
2022.07
DiffRF[63]
CVPR 2022.12
VGOS[39]
IJCAI 2023.04
NSVF[19]
NeurIPS 2020.07
Mip-NeRF[14]
ICCV 2021.03
Ref-NeRF[16]
CVPR 2021.12
HumanNeRF[74]
CVPR 2022.01
DreamFusion[58]
ICLR 2022.09
PAniC-3D[75]
CVPR 2023.03

3D Gaussian Splatting

3D Gaussian Splatting for Real-Time Radiance Field Rendering

BERNHARD KERBL*, Inria, Université Côte d'Azur, France
GEORGIOS KOPANAS*, Inria, Université Côte d'Azur, France
THOMAS LEIMKÜHLER, Max-Planck-Institut für Informatik, Germany
GEORGE DRETTAKIS, Inria, Université Côte d'Azur, France

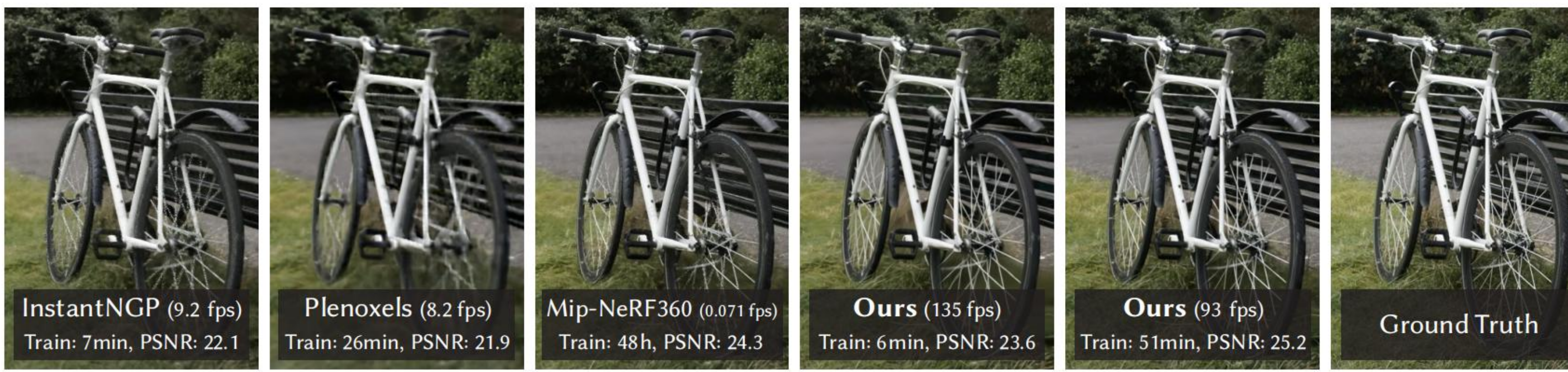


Fig. 1. Our method achieves real-time rendering of radiance fields with quality that equals the previous method with the best quality [Barron et al. 2022], while only requiring optimization times competitive with the fastest previous methods [Fridovich-Keil and Yu et al. 2022; Müller et al. 2022]. Key to this performance is a novel 3D Gaussian scene representation coupled with a real-time differentiable renderer, which offers significant speedup to both scene optimization and novel view synthesis. Note that for comparable training times to InstantNGP [Müller et al. 2022], we achieve similar quality to theirs; while this is the maximum quality they reach, by training for 51min we achieve state-of-the-art quality, even slightly better than Mip-NeRF360 [Barron et al. 2022].

ACM TOG 2023

3D Gaussian Splatting

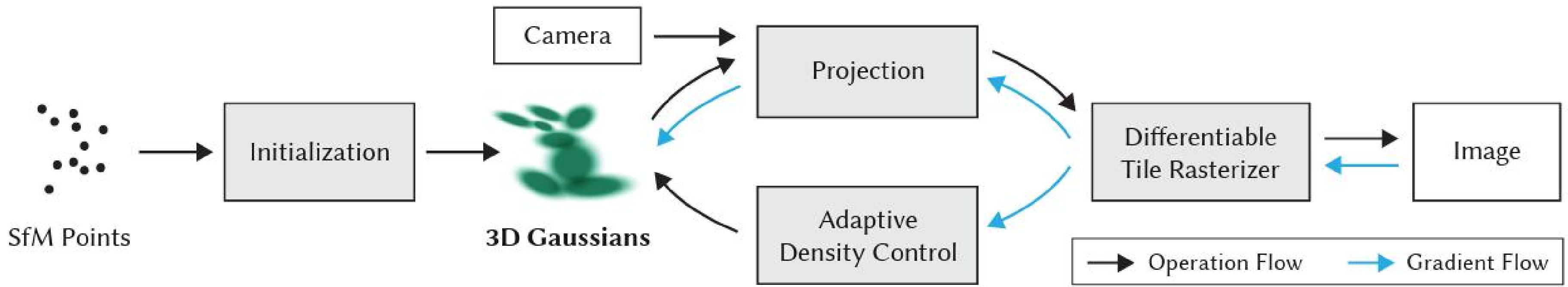


Fig. 2. Optimization starts with the sparse SfM point cloud and creates a set of 3D Gaussians. We then optimize and adaptively control the density of this set of Gaussians. During optimization we use our fast tile-based renderer, allowing competitive training times compared to SOTA fast radiance field methods. Once trained, our renderer allows real-time navigation for a wide variety of scenes.

ACM TOG 2023

This video contains a voice-over

3D Gaussian Splatting for Real-Time Radiance Field Rendering

SIGGRAPH 2023
(ACM Transactions on Graphics)

Bernhard Kerbl* — Inria, Université Côte d'Azur
Georgios Kopanas* — Inria, Université Côte d'Azur
Thomas Leimkühler — max planck institut informatik
George Drettakis — Inria, Université Côte d'Azur

* Denotes equal contribution

GraphDeco
GRAPHics and Design with hEterogeneous COntent

Geometric Deep Learning

Geometric Deep Learning
Grids, Groups, Graphs,
Geodesics, and Gauges

Michael M. Bronstein[1], Joan Bruna[2], Taco Cohen[3], Petar Veličković[4]

May 4, 2021

PointNet

PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation

Charles R. Qi, Hao Su, Kaichun Mo, Leonidas J. Guibas; Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 652-660

Abstract

Point cloud is an important type of geometric data structure. Due to its irregular format, most researchers transform such data to regular 3D voxel grids or collections of images. This, however, renders data unnecessarily voluminous and causes issues. In this paper, we design a novel type of neural network that directly consumes point clouds, which well respects the permutation invariance of points in the input. Our network, named PointNet, provides a unified architecture for applications ranging from object classification, part segmentation, to scene semantic parsing. Though simple, PointNet is highly efficient and effective. Empirically, it shows strong performance on par or even better than state of the art. Theoretically, we provide analysis towards understanding of what the network has learnt and why the network is robust with respect to input perturbation and corruption.

PointNet

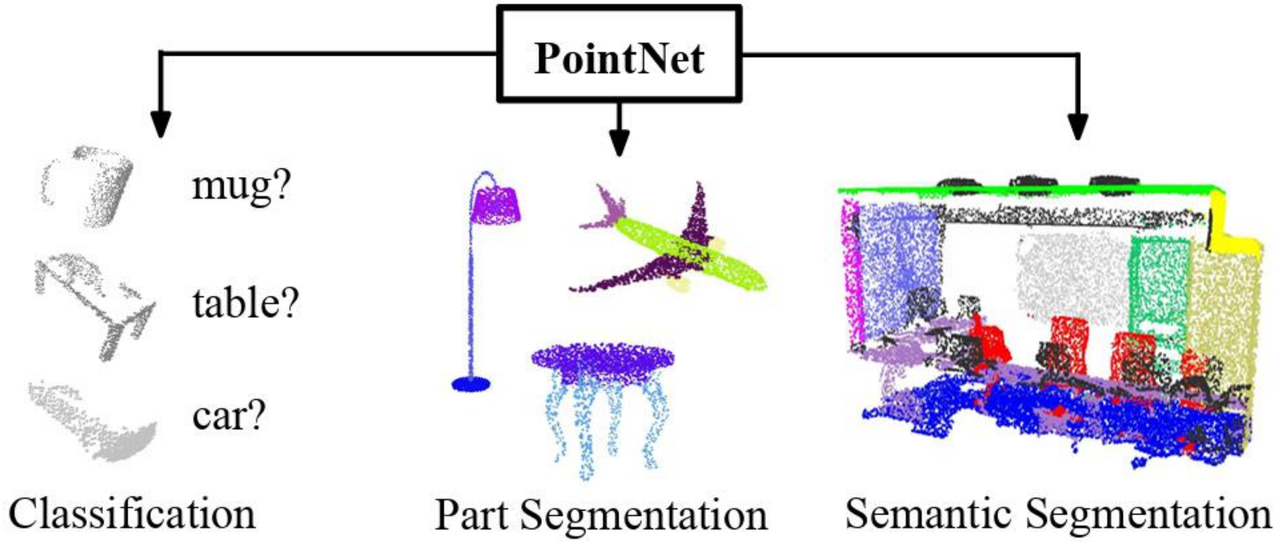


Figure 1. **Applications of PointNet.** We propose a novel deep net architecture that consumes raw point cloud (set of points) without voxelization or rendering. It is a unified architecture that learns both global and local point features, providing a simple, efficient and effective approach for a number of 3D recognition tasks.

PointNet

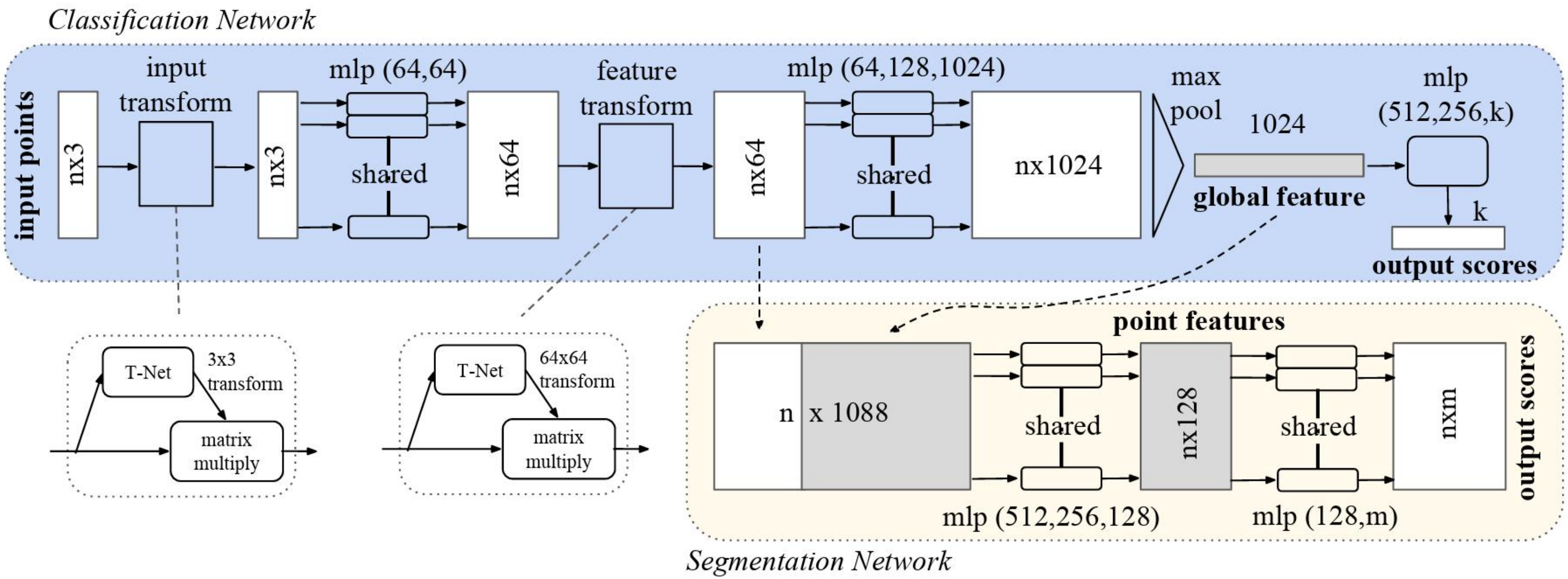


Figure 2. **PointNet Architecture.** The classification network takes n points as input, applies input and feature transformations, and then aggregates point features by max pooling. The output is classification scores for k classes. The segmentation network is an extension to the classification net. It concatenates global and local features and outputs per point scores. "mlp" stands for multi-layer perceptron, numbers in bracket are layer sizes. Batchnorm is used for all layers with ReLU. Dropout layers are used for the last mlp in classification net.

Deep Motion Editing

A Deep Learning Framework for Character Motion Synthesis and Editing

Daniel Holden*
University of Edinburgh

Jun Saito†
Marza Animation Planet

Taku Komura‡
University of Edinburgh

Figure 1: *Our framework allows the animator to synthesize character movements automatically from given trajectories.*

siggraph 2016

Deep Motion Editing

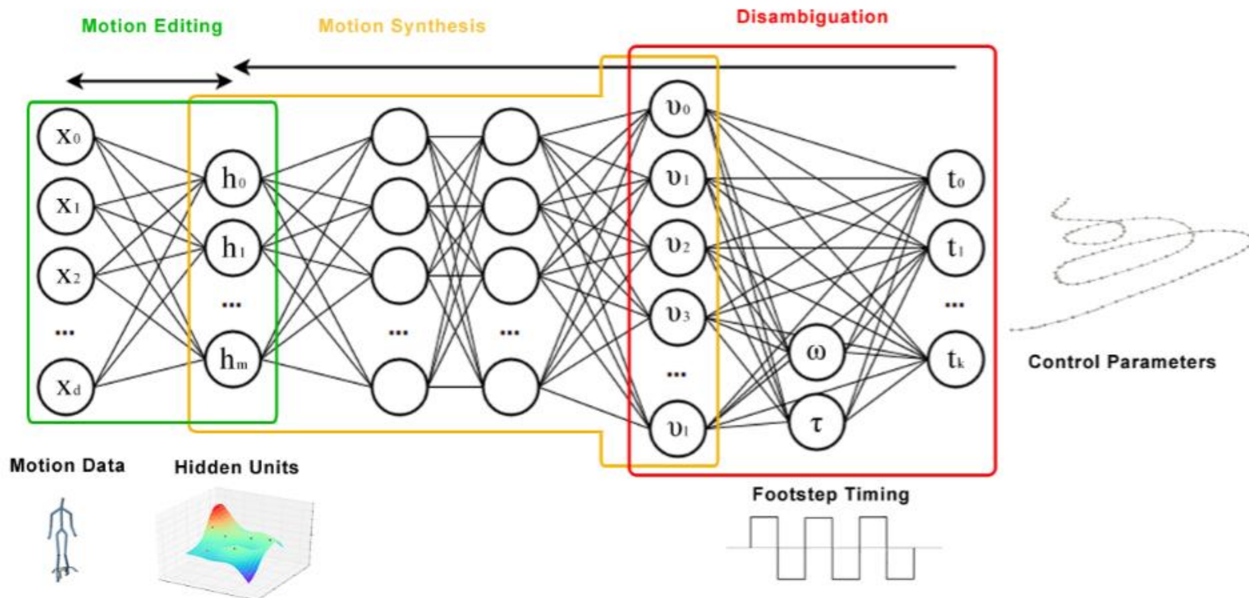


Figure 2: *The outline of our method. High level parameterizations are disambiguated and used as input to feed forward neural networks. These networks produce motion in the space of the hidden units of a convolutional autoencoder which can be further used to edit the generated motion.*

siggraph 2016

Text2Motion

Generating Diverse and Natural 3D Human Motions from Text

Chuan Guo   Shihao Zou   Xinxin Zuo   Sen Wang   Wei Ji   Xingyu Li
Li Cheng
University of Alberta
{cguo2, szou2,wji3,xingyu,lcheng5}@ualberta.ca

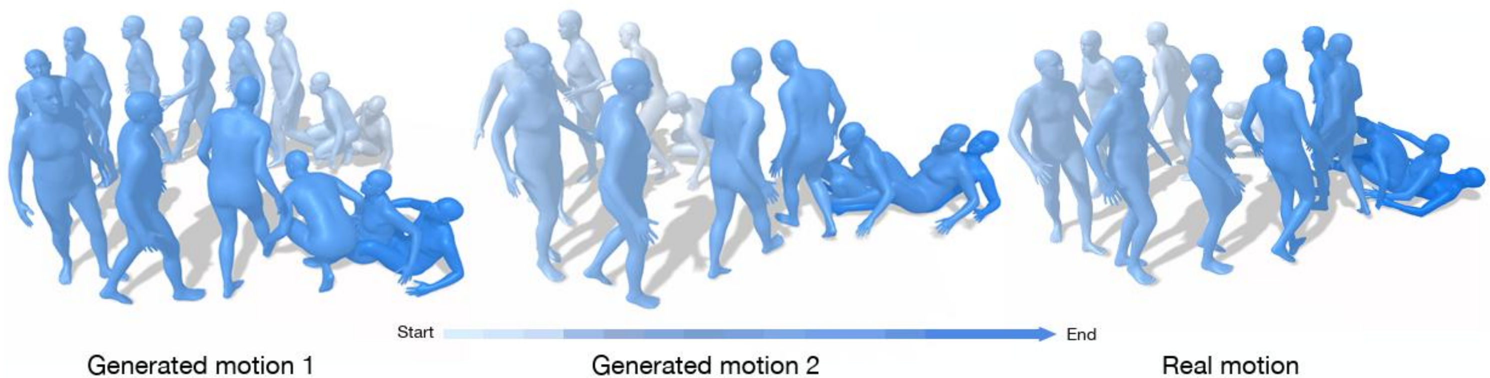
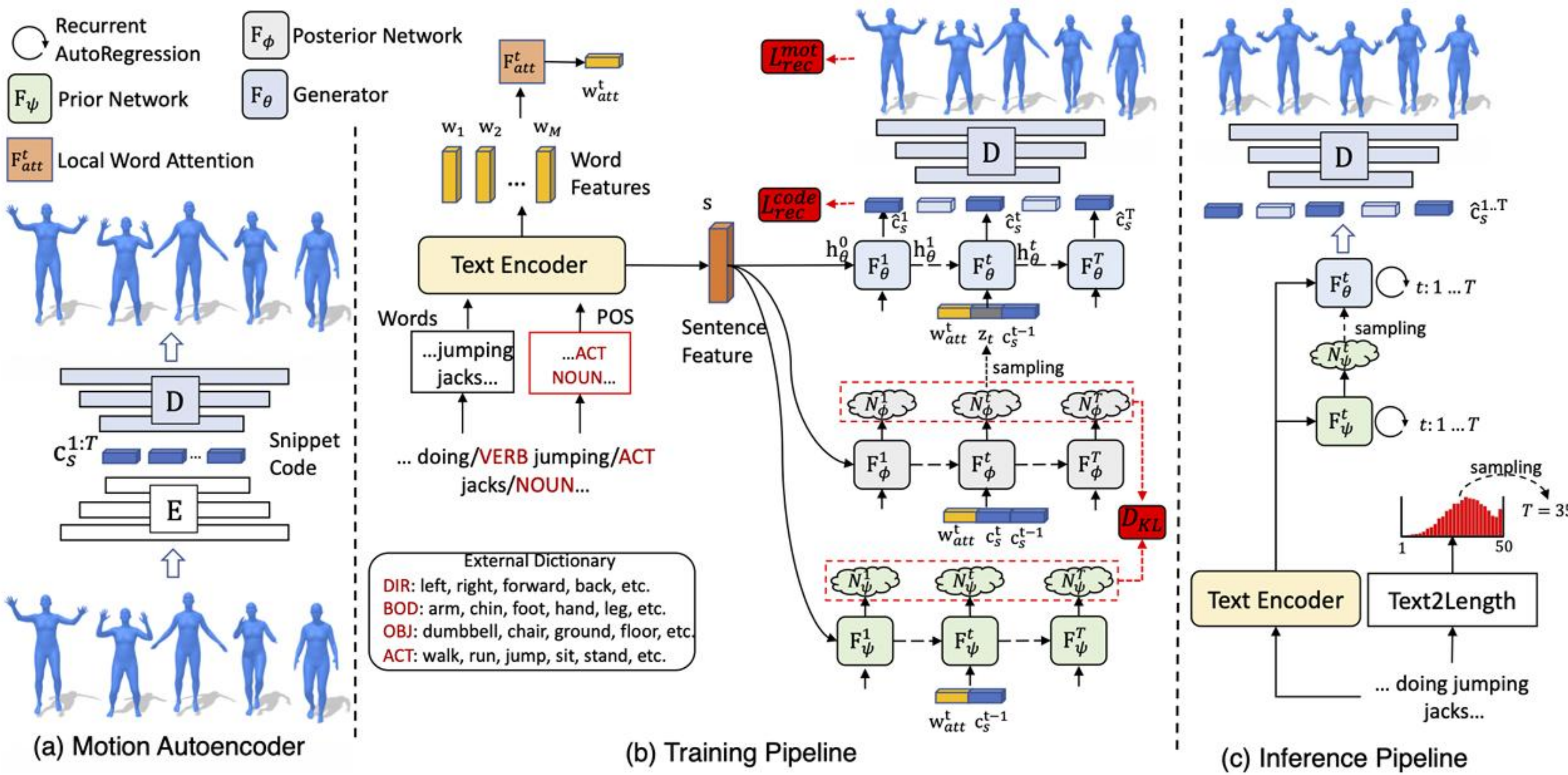


Figure 1. Taken as input the text description, *"the figure rises from a lying position and walks in a counterclockwise circle, and then lays back down the ground"*, our approach generates multiple distinct 3D human motions (e.g. the left and middle panels) that are faithful to the prescribed textual content. The real motion is also presented at the right panel for reference.

cvpr 2022

Text2Motion



cvpr 2022

Generating Diverse and Natural 3D Motions from Text

Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, Cheng Li

University of Alberta

Facial Capture w DCNN

Production-Level Facial Performance Capture Using Deep Convolutional Neural Networks

Samuli Laine
NVIDIA

Tero Karras
NVIDIA

Timo Aila
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Antti Herva
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sca 2017

Facial Capture w DCNN

ABSTRACT

We present a real-time deep learning framework for video-based facial performance capture—the dense 3D tracking of an actor's face given a monocular video. Our pipeline begins with accurately capturing a subject using a high-end production facial capture pipeline based on multi-view stereo tracking and artist-enhanced animations. With 5–10 minutes of captured footage, we train a convolutional neural network to produce high-quality output, including self-occluded regions, from a monocular video sequence of that subject. Since this 3D facial performance capture is fully automated, our system can drastically reduce the amount of labor involved in the development of modern narrative-driven video games or films involving realistic digital doubles of actors and potentially hours of animated dialogue per character. We compare our results with several state-of-the-art monocular real-time facial capture techniques and demonstrate compelling animation inference in challenging areas such as eyes and lips.

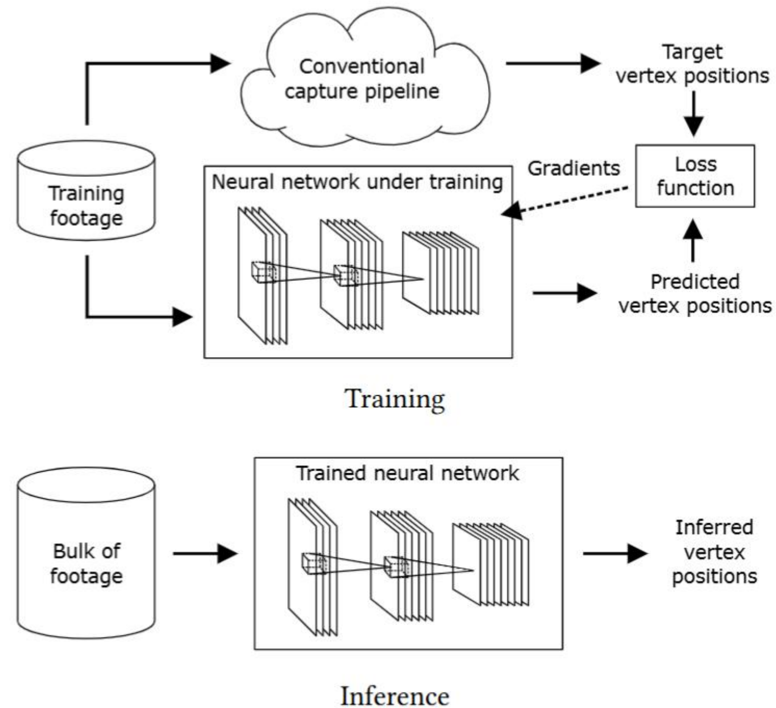

Figure 1: Our deep learning-based facial performance capture framework is divided into a training and inference stage. The goal of our system is to reduce the amount of footage that needs to be processed using labor-intensive production-level pipelines.

sca 2017

Production-Level Facial Performance Capture Using Deep Convolutional Neural Networks

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NVIDIA

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Audio-driven Facial Animation

Audio-Driven Facial Animation by Joint End-to-End Learning of Pose and Emotion

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TIMO AILA, NVIDIA
SAMULI LAINE, NVIDIA
ANTTI HERVA, Remedy Entertainment
JAAKKO LEHTINEN, NVIDIA and Aalto University

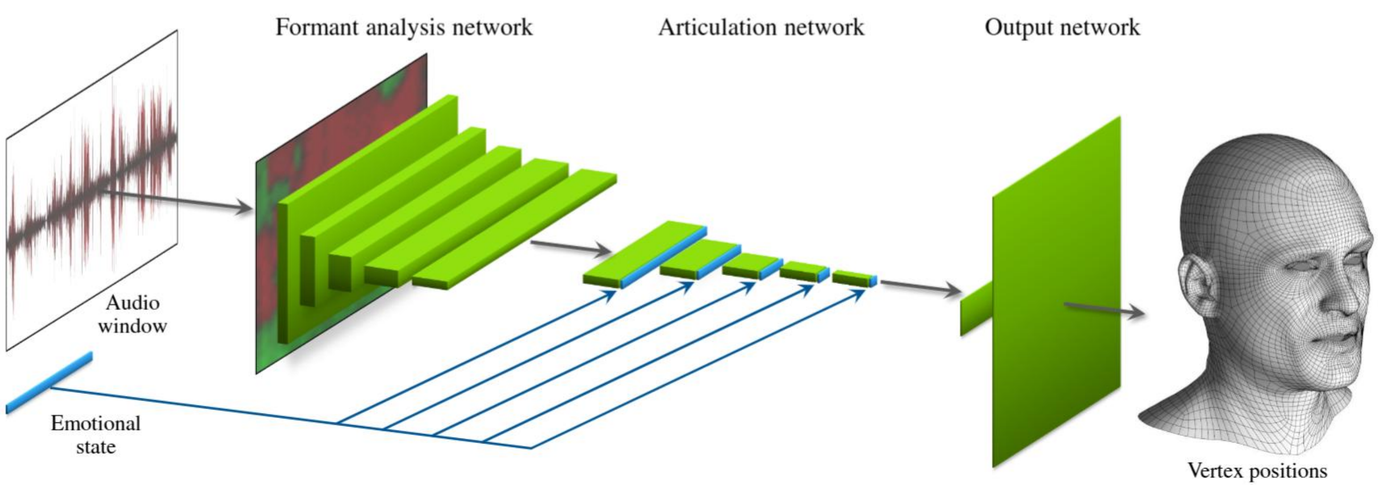

Fig. 1. Our deep neural network for inferring facial animation from speech. The network takes approximately half a second of audio as input, and outputs the 3D vertex positions of a fixed-topology mesh that correspond to the center of the audio window. The network also takes a secondary input that describes the emotional state. Emotional states are learned from the training data without any form of pre-labeling.

acm tog 2017

Audio-Driven Facial Animation by Joint End-to-End Learning of Pose and Emotion

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Samuli Laine	NVIDIA
Antti Herva	Remedy Entertainment
Jaakko Lehtinen	NVIDIA, Aalto University

Video-driven Facial Animation Transfer

Semi-supervised video-driven facial animation transfer for production

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CHINYU CHIEN, Digital Domain, Taiwan
MARK WILLIAMS and JOSE SERRA, Digital Domain, Canada
DARREN HENDLER and DOUG ROBLE, Digital Domain, USA

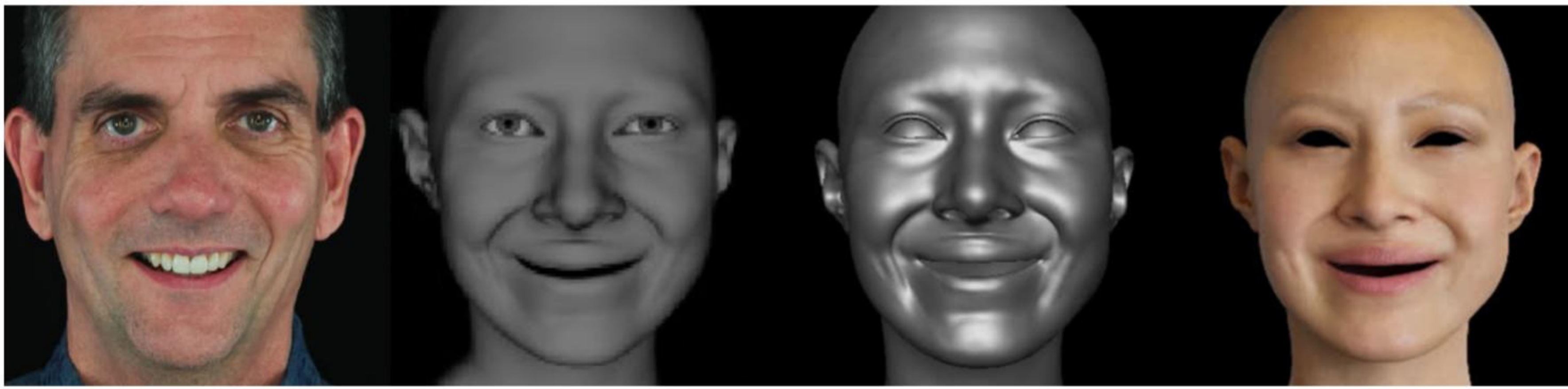

Fig. 1. Results of our image-to-geometry method. From left-to-right: aligned input image, translated character image, our output geometry and high-quality render of the output.

siggraph Asia 2021

Video-driven Facial Animation Transfer

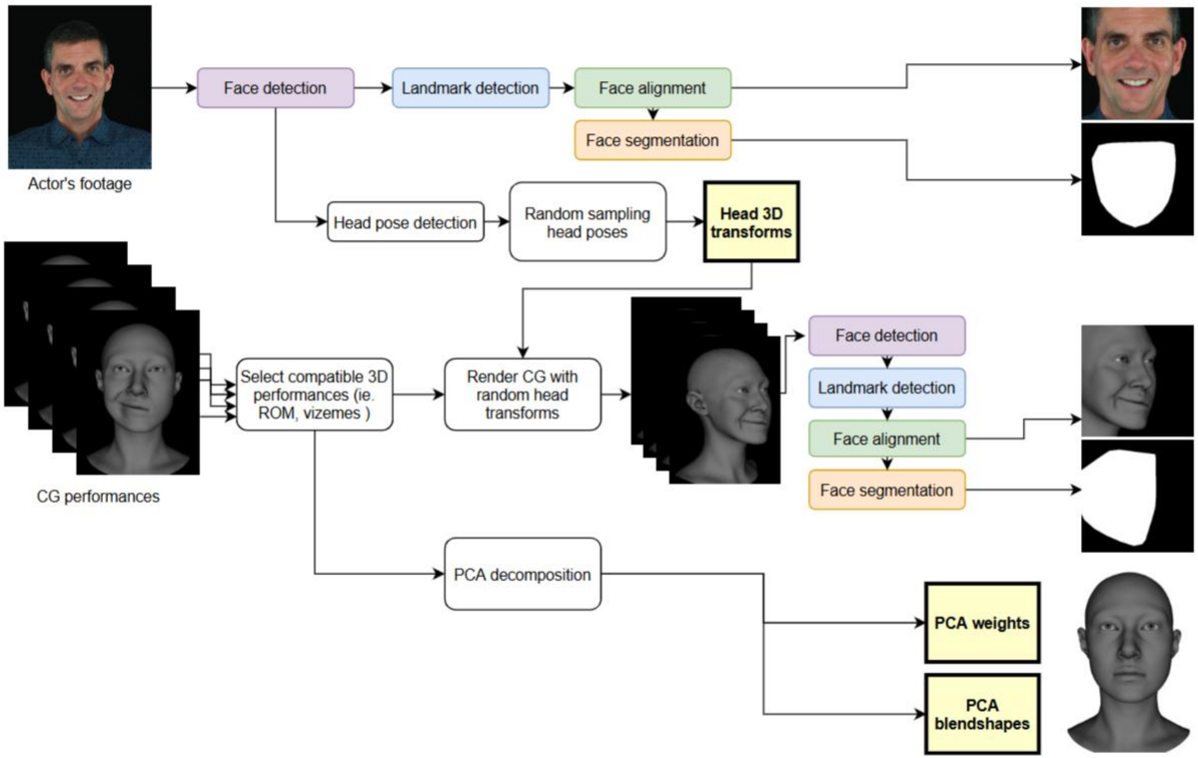

Fig. 2. Data preparation steps: Real footage goes through face detection, landmark detection, face alignment and face segmentation to generate a sequence of aligned face images and background masks. Character dataset is determined based on compatible CG performances to the real footage, then rendered using head poses randomly sampled from the distribution obtained from the real footage. The rendered images follow the same image processes to derive the aligned CG face images and background masks. Additionally, the selected geometry is linearly decomposed with PCA into weights and blendshapes.

siggraph Asia 2021

Video-driven Facial Animation Transfer

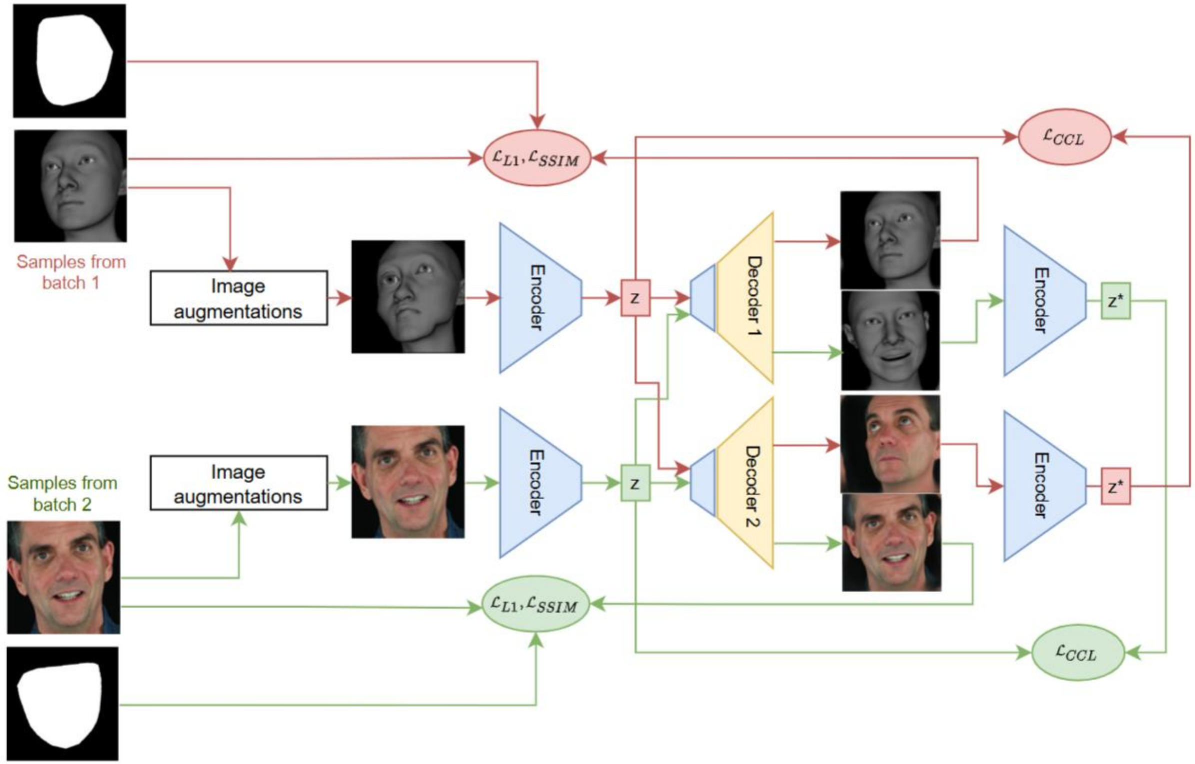

Fig. 3. Structure to train image-to-image translation network for two domains. Showing two autoencoders with shared components in blue and domain-specific components in yellow. The shared encoder is depicted multiple times to illustrate the symmetrical computation for the loss terms in the two consecutive batches (green and red arrows). For each domain, the aligned face images are augmented and fed to the autoencoders. L1 and SSIM image loss terms enforce that the masked output images match the masked input images. A cross domain latent cycle-consistency loss (CCL) term encourages the shared encoder to map the latent codes z, computed from the input images, to similar latent codes z^*, computed from the decoded images of the opposite domain.

siggraph Asia 2021

SIGGRAPH ASIA 2021 TOKYO

Semi-supervised video-driven facial animation transfer for production

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Jose Serra
CANADA

Darren Hendler
USA

Doug Roble
USA

DIGITAL DOMAIN

Real-time Expression Transfer

Real-time Expression Transfer for Facial Reenactment

Justus Thies[1] Michael Zollhöfer[2] Matthias Nießner[3] Levi Valgaerts[2] Marc Stamminger[1] Christian Theobalt[2]
[1]University of Erlangen-Nuremberg [2]Max-Planck-Institute for Informatics [3]Stanford University

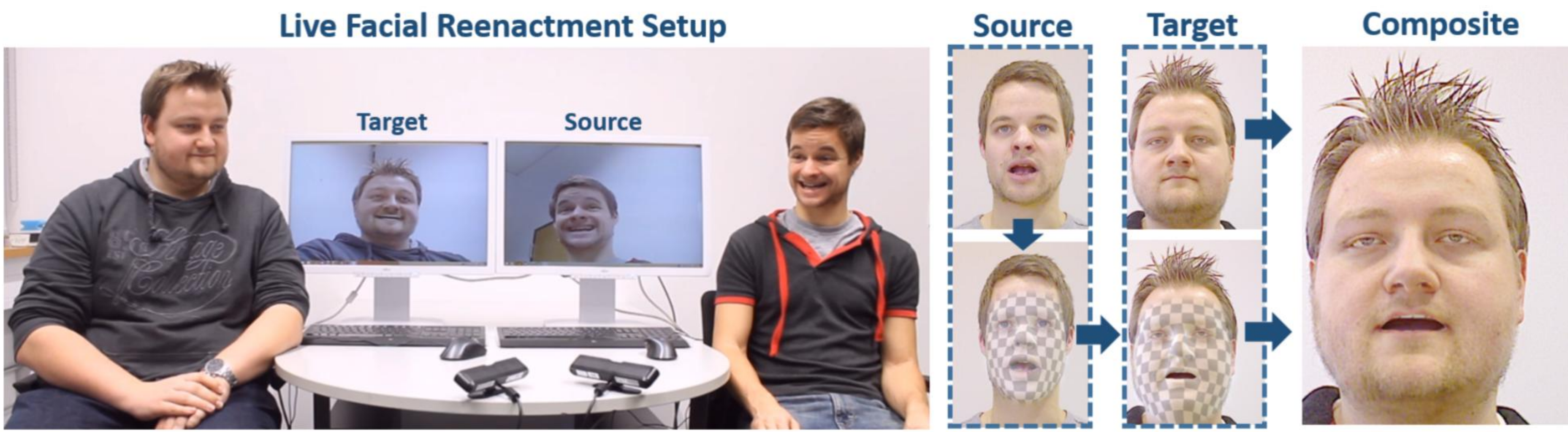


Figure 1: *Our live facial reenactment technique tracks the expression of a source actor and transfers it to a target actor at real-time rates. The synthetic result is photo-realisticly re-rendered on top of the original input stream maintaining the target's identity, pose and illumination.*

acm tog 2015

Real-time Expression Transfer

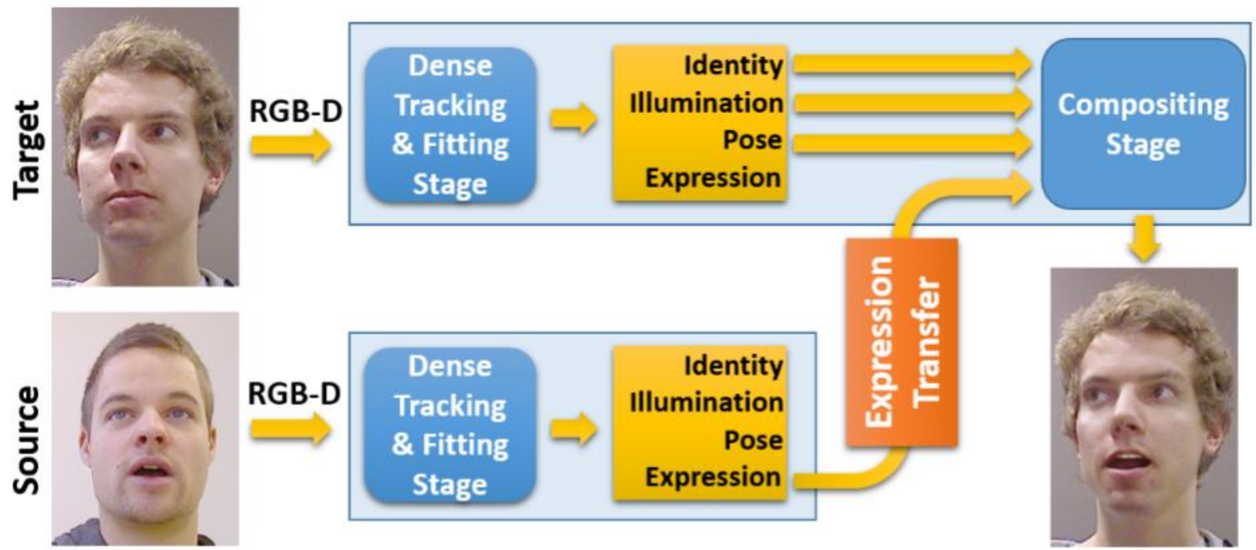


Figure 2: *Our live facial reenactment pipeline.*

acm tog 2015

Real-time Expression Transfer for Facial Reenactment

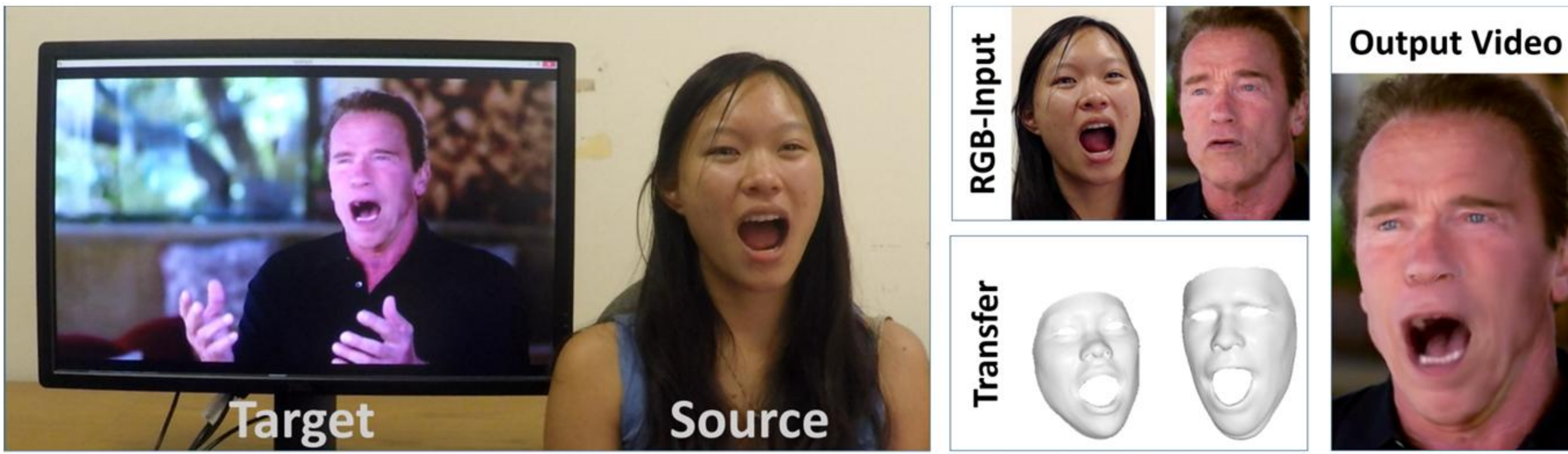
Justus Thies[1], *Michael Zollhöfer*[2], *Matthias Nießner*[3], *Levi Valgaerts*[2], *Marc Stamminger*[1], *Christian Theobalt*[2]

[1]University of Erlangen-Nuremberg
[2]Max-Planck-Institute for Informatics
[3]Stanford University

Face2Face

Face2Face: Real-time Face Capture and Reenactment of RGB Videos

Justus Thies[1] Michael Zollhöfer[2] Marc Stamminger[1] Christian Theobalt[2] Matthias Nießner[3]
[1]University of Erlangen-Nuremberg [2]Max-Planck-Institute for Informatics [3]Stanford University



Proposed online reenactment setup: a monocular target video sequence (e.g., from Youtube) is reenacted based on the expressions of a source actor who is recorded live with a commodity webcam.

cvpr 2016

Face2Face

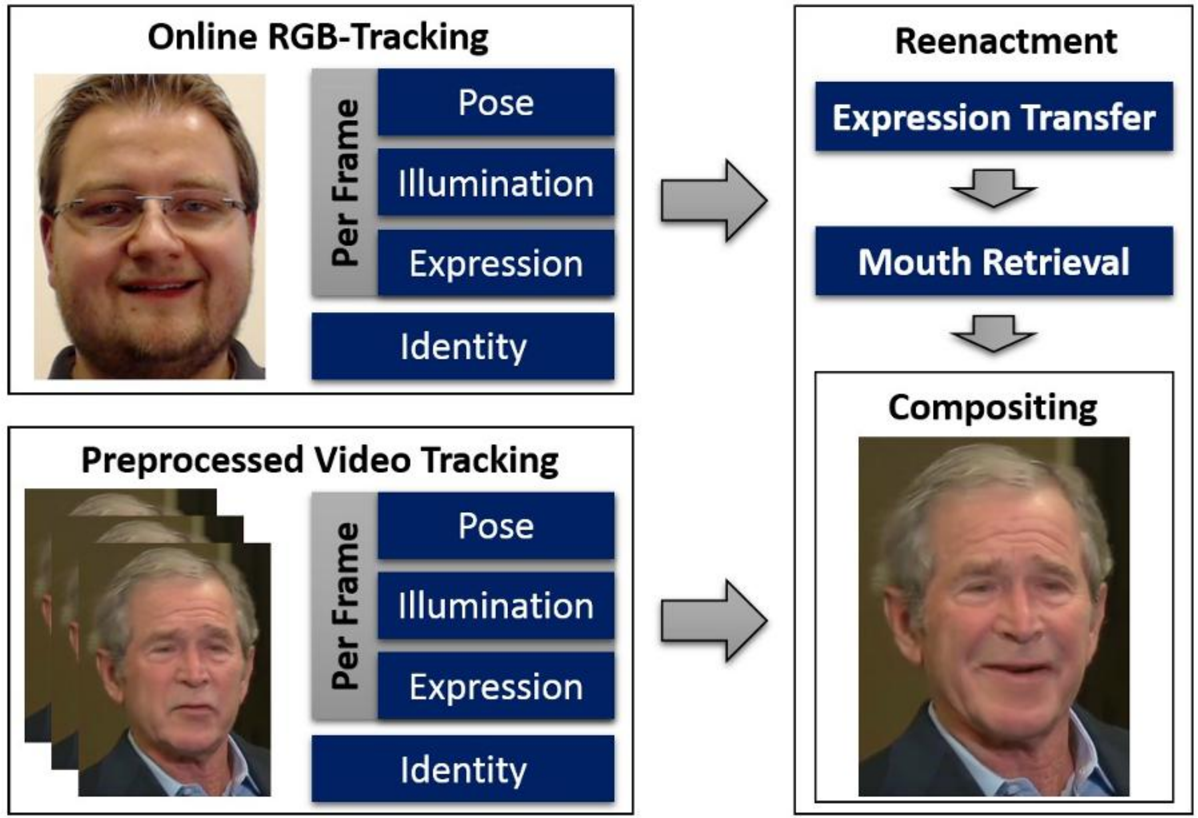


Figure 1: Method overview.

cvpr 2016

Face2Face: Real-time Face Capture and Reenactment of RGB Videos

Justus Thies[1], *Michael Zollhöfer*[2],
Marc Stamminger[1], *Christian Theobalt*[2],
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CVPR 2016 (Oral)

One-Shot Audio-Driven 3D Talking Face Generation by Real3D-Portrait

<Real3D-Portrait: One-shot Realistic 3D Talking Portrait Synthesis>

- Real3D-Portrait is the **first one-shot audio-driven 3D method** that could synthesize realistic **3D talking face** videos given only **one source image** and the **driving audio signals**. It features in:
  1. Generalized reconstruction of 3D head and accurate face animation;
  2. Natural torso and background segments;
  3. Cross-lingual-generalized and accurate audio-lip synchornization.

- In this video, we test 10 unseen identities driven by 6 audio clips in this video, including:
  1. Male English (15 seconds);
  2. Female English (10 seconds);
  3, French (11 seconds);
  4. Japanese (9 seconds);
  5. Spanish (15 seconds);
  6. Korean (33 seconds).
- We recommend the viewer to evaluate the generated videos from the aspect of "*identity preserving*", "*Visual Quality*", and "*audio-lip synchronization*".

EMO: Emote Portrait Alive - Generating Expressive Portrait Videos with Audio2Video Diffusion Model under Weak Conditions

LinRui Tian, Qi Wang, Bang Zhang, LieFeng Bo
Institute for Intelligent Computing, Alibaba Group

其他

ICARUS: A Specialized Architecture for Neural Radiance Fields Rendering

CHAOLIN RAO, ShanghaiTech University, China and GGU Technology Co., Ltd., China
HUANGJIE YU, ShanghaiTech University, China
HAOCHUAN WAN, ShanghaiTech University, China
JINDONG ZHOU, ShanghaiTech University, China
YUEYANG ZHENG, ShanghaiTech University, China
MINYE WU, KU Leuven, Belgium
YU MA, ShanghaiTech University, China
ANPEI CHEN, ShanghaiTech University, China
BINZHE YUAN, ShanghaiTech University, China
PINGQIANG ZHOU, ShanghaiTech University, China
XIN LOU*, ShanghaiTech University, China and GGU Technology Co., Ltd., China
JINGYI YU*, ShanghaiTech University, China

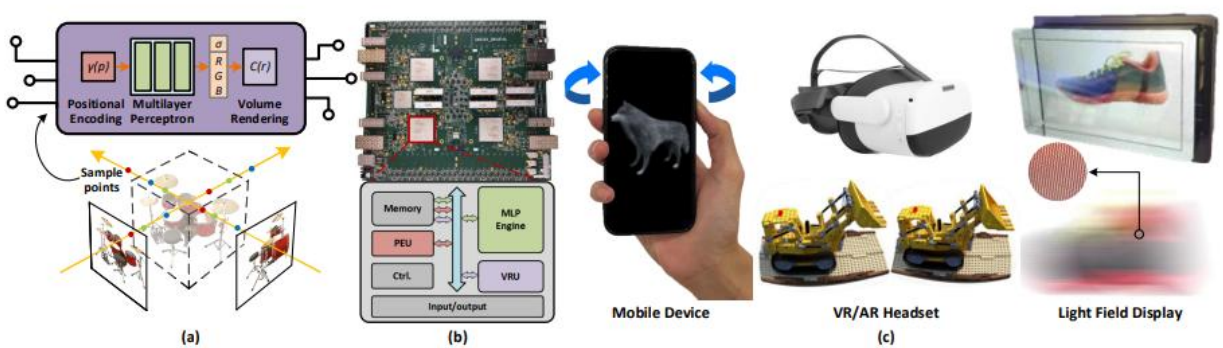

Fig. 1. We demonstrate a specialized hardware architecture for NeRF-based rendering applications. (a) Our hardware design involves operations in NeRF-based rendering. (b) We validate our architecture using an FPGA platform. (c) Potential applications that can benefit from our architecture.

讨论：AI4CG？CG4AI？

- NERF, 3D Gaussian Splatting
- G-Deeplearning, PointNet
- DeepMotionEditing, text2motion
- DeepFake, Video Generation

探讨

AI会给计算机图形学带来何种影响?

拓展

https://www.realtimerendering.com/kesen/sig2025.html

https://www.realtimerendering.com/kesen/sig2023.html

https://www.realtimerendering.com/kesen/sig2020.html

结语

专业普及课；

主要讲授计算机图形学领域的主要概念、算法思想和基本技术；

涉及几何计算、渲染、动画模拟、三维重建与运动捕捉等；

了解计算机图形学的前沿研究领域和最新研究成果。

谢谢！欢迎交流！